

Sampling through the witness structure

A one-page companion to Feng–Jin, “A Faster Algorithm for Counting Knapsack Solutions” (SODA 2025, arXiv:2410.22267), whose notation and numbering are used throughout.

Where their time goes. The total $\tilde{O}(n^{1.5}\varepsilon^{-2})$ splits into *construction* — the merge trees of Theorem 4.2, $\tilde{O}(n^{1.5})$ — and *queries*: $N = \tilde{O}(n/(\ell\varepsilon^2))$ samples (Lemma 3.3), each answered by descending the trees and drawing, at every node, a split from a distribution of support $\tilde{O}(\ell)$ by scanning it. The claim here: the query stage can run in nearly the time it takes to write the samples down, giving $\tilde{O}(n^{1.5} + \min(n\sqrt{\ell}, n^2/\ell) \cdot \varepsilon^{-2}) \leq \tilde{O}(n^{1.5} + n^{4/3}\varepsilon^{-2})$ — never worse than $\tilde{O}(n^{1.5}\varepsilon^{-2})$, and below it for every $\varepsilon = o(1)$.

The identity. Recall the merge of §6.2: levels $A_0^*[x]$ with $f_0(x) \in [D^{A_0^*[x]}, D^{A_0^*[x]+1})$, prefix maxima $A[x]$, residuals $u[x] = f_0(x)D^{-A[x]} \in [1, D)$ at prefix-max-attaining positions (else 0); likewise B, v for g_0 ; $C =$ the $(\max, +)$ -convolution of A, B ; $w[z] = \sum_{x: A[x]+B[z-x]=C[z]} u[x]v[z-x]$ (Theorem 6.1); the merge returns $h(z) = D^{C[z]} \cdot w[z]$. Each witness term satisfies $u[x]v[z-x]D^{C[z]} = f_0(x)g_0(z-x)$ *exactly* — the exponents cancel by the witness condition. Hence

$$h(z) = \sum \{ f_0(x)g_0(y) : x+y=z, A[x]+B[y]=C[z], u[x]v[y]>0 \}.$$

So the stored array is not merely a $(1 \pm \delta)$ -approximation of $f \star g$: it is an *exact aggregation* of the product measure restricted to witness pairs (a measure that is within a per-prefix $(1 \pm \delta)$ of $f \star g$ by (41)). The construction, built to *compute* the arrays, stores for free an exact factorization of the distribution they define.

The sampler. Sample that factorization literally. (i) At the root: a coarse value $z \propto \hat{f}_{\text{root}}(z) \cdot \mathbf{1}[z \leq t]$ — the capacity appears here and nowhere else. (ii) De-rounding: a node’s fine position $s \propto h(s)$ over the preimage of its coarse value; position rounding is a monotone floor, so the preimage is a contiguous interval — an $O(\log)$ prefix-sum draw, no differences of approximate quantities anywhere. (iii) Split: a witness pair $(x, y = s - x) \propto u[x]v[y]$ over $\{A[x] + B[y] = C[s]\}$; recurse with the *exact* child values x, y . Every draw is exactly proportional to the stored decomposition, so the sampled measure is the one the arrays define, and the TV analysis of §4–5 applies unchanged; the only new TV term is the output re-rounding of h (a pointwise $1 \pm \delta$, so $\leq 2\delta$ per visited merge). Children with value 0 whose weight-0 stratum is only $\emptyset$ are not descended, so a sample visits only the path-union of its k items.

Why the split is cheap. Witness pairs at s organize by level: $a + b = C[s]$, and the attaining positions of A -level a form an interval $I^{(a)}$ (A is monotone), likewise $J^{(b)}$ — at most $M = \tilde{O}(\text{subtree items}/\log D)$ rectangles, against array length L ; near the roots $M \ll L$. Within a rectangle, $u, v \in [1, D)$ have $O(\log D)$ log-range: split into factor-2 classes X_i, Y_j ; the counts $|X_i \cap (s - Y_j)|$ for *all* s are indicator convolutions, FFT-precomputed in $\tilde{O}(L \text{ polylog})$ per node. A draw picks (rectangle, class) $\propto 2^{i+j} \cdot \text{count}$, a uniform member by rank (prefix counts from lazily built, memoized dyadic restrictions; unconditionally $\tilde{O}(L\sqrt{N})$ amortized per node), then accepts with probability $u[x]v[y]/2^{i+j+2} \geq \frac{1}{4}$ — exact. Diagonal pairs *below* $C[s]$ carry $\leq \delta$ of the diagonal mass by the multiples-of-3 separation, exactly as in (41): one more $O(1)$ rejection.

The count. Per split-draw $\tilde{O}(\min(M, L))$; per sample, over the pruned path-union, $\tilde{O}(\min(n_m, \ell\sqrt{k}))$; over $N = \tilde{O}(n/(\ell\varepsilon^2))$ samples, $\tilde{O}(\min(n^2/\ell, n\sqrt{\ell}) \cdot \varepsilon^{-2}) \leq \tilde{O}(n^{4/3}\varepsilon^{-2})$, the maximum at $\ell = n^{2/3}$. Construction is unchanged; the second phase (§5.4) admits the same treatment.

Status. The identity, the δ -domination, the lazy amortization, and the min-collapse are machine-checked in Lean (`lean/SharpKnapsack/WitnessSampler.lean`); every sampling mechanism is validated against brute force (`experiments/test_witness_sampler.py`). This is a *theorem-candidate*: what remains is the substitution pass through Lemmas 4.6/4.7 and §5.3–5.4 and the detailed second-phase ledger (`docs/witness-sampler.md`).